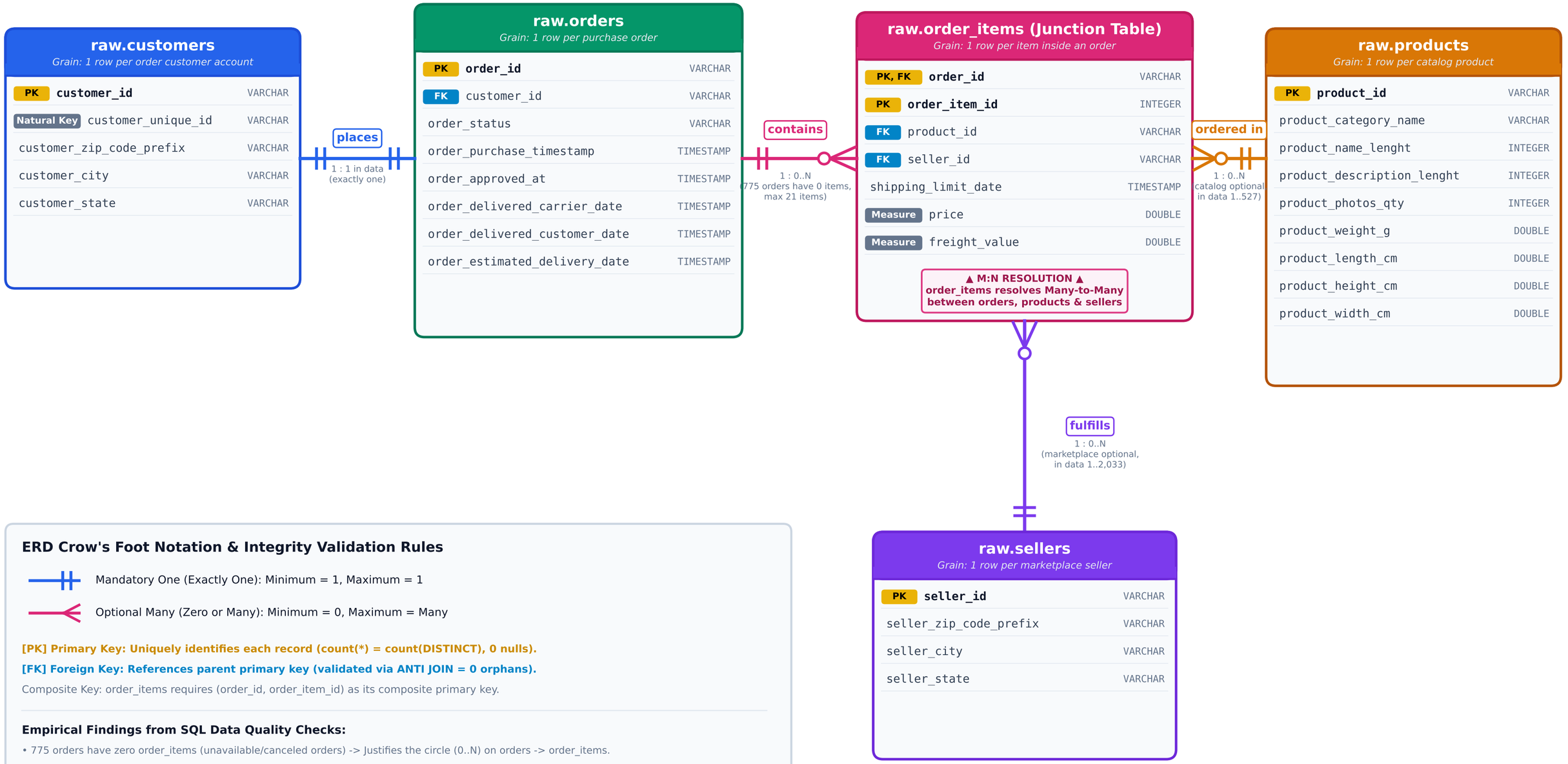
BITS College • Department of Software Engineering • SWEN466: Knowledge Discovery & Data Mining

Olist E-Commerce Entity-Relationship Diagram (Crow's Foot Notation)

Primary Keys [PK], Foreign Keys [FK], Cardinalities & Relationship Verbs



ERD Crow's Foot Notation & Integrity Validation Rules

Mandatory One (Exactly One): Minimum = 1, Maximum = 1

Optional Many (Zero or Many): Minimum = 0, Maximum = Many

[PK] Primary Key: Uniquely identifies each record (count(*) = count(DISTINCT), 0 nulls).

[FK] Foreign Key: References parent primary key (validated via ANTI JOIN = 0 orphans).

Composite Key: order_items requires (order_id, order_item_id) as its composite primary key.

Empirical Findings from SQL Data Quality Checks:

- 775 orders have zero order_items (unavailable/canceled orders) -> Justifies the circle (0..N) on orders -> order_items.
- In the raw data, every product and seller has at least 1 order_item (min = 1) -> Data snapshot is stricter than catalog diagram (1..N vs 0..N).
- customer_id is created per purchase transaction (1:1 with orders); customer_unique_id identifies the real recurring human.
- Zero orphaned child records: 100% referential integrity across all child-to-parent foreign key relationships.